

AI-Driven Smart City Service Management using Predictive Analytics and Reinforcement Learning

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Abstract—Municipal 311 systems typically allocate service crews after requests are received, limiting their ability to anticipate spatial variations in demand. This paper presents a simulation-based, two-stage framework for proactive crew allocation across Vancouver’s local areas. In the first stage, an XGBoost model forecasts area-level demand using historical service-request data. In the second stage, a Proximal Policy Optimization agent combines multi-day demand forecasts with current backlog, the age of unresolved requests, and crew-location information to prioritize service zones. A continuous priority-vector formulation is introduced to make the multi-crew allocation problem computationally tractable. Experimental evaluation against three reactive baseline policies shows that the proposed approach achieves better overall performance and greater consistency under sustained capacity constraints. These findings demonstrate the potential of combining predictive analytics and reinforcement learning to support proactive and efficient municipal service management.

Index Terms—smart city, reinforcement learning, proximal policy optimization, demand forecasting, XGBoost, resource allocation, crew scheduling, urban service management, predictive analytics, Markov decision process

I. INTRODUCTION

Municipal 311 systems manage non-emergency requests involving infrastructure maintenance, sanitation, public spaces, and other city services. These systems typically allocate crews after requests are received, limiting their ability to anticipate spatial and temporal variations in demand. Although this reactive approach supports request tracking and dispatch, it may contribute to growing backlogs, uneven service levels across city areas, and inefficient use of limited crew capacity when demand fluctuates or exceeds available resources. Previous studies have shown that municipal service demand contains predictable spatial and temporal patterns that can support operational planning [1], [2].

This paper presents a two-stage framework for proactive municipal service management. In the first stage, historical 311 service-request data from the City of Vancouver [3] are used to train an XGBoost demand-forecasting model [4]. The model produces daily demand forecasts for Vancouver’s 22 local areas, each of which is treated as a service zone. In the second stage, a reinforcement learning agent trained using Proximal Policy Optimization (PPO) [5] combines a five-day demand forecast with current backlog, the age of unresolved

requests, and crew-location information to prioritize service zones. This enables the agent to consider anticipated demand when making daily allocation decisions.

A central challenge is the size of the multi-crew assignment space. If each of 11 crews independently selects one of 22 zones, the direct joint action space contains $22^{11} \approx 5.84 \times 10^{14}$ possible labeled assignments. To make the problem computationally tractable, the proposed policy outputs a continuous 22-dimensional priority vector. The 11 highest-ranked zones are then selected, with one crew assigned to each selected zone. This formulation reduces the complexity of the policy output while enforcing allocation to distinct service zones. Because this constraint differs from unrestricted multi-crew assignment, the proposed formulation represents a structured version of the original allocation problem.

The main contributions of this paper are as follows:

- A two-stage framework that integrates area-level demand forecasting with reinforcement learning for proactive municipal crew allocation.
- A custom simulation environment that models daily demand, backlog accumulation, backlog age, crew movement, service capacity, and travel costs.
- A continuous priority-vector formulation that makes multi-zone crew allocation tractable while ensuring that the selected crews are distributed across distinct service zones.
- An empirical evaluation against three baseline policies—Magnitude-Based Greedy, Reactive FCFS, and Oldest-or-Closest—under paired simulated demand conditions.

II. RELATED WORK

Forecasting municipal service demand is an important challenge in urban operations. Xu et al. [1] developed an adaptive space-time kernel method for forecasting 311 requests, demonstrating predictable spatial and temporal patterns. Cheng et al. [2] compared machine learning models, including Random Forest and XGBoost, for predicting request volumes across Miami-Dade County. Although these studies support demand forecasting and resource planning, they do not address sequential crew allocation under limited service capacity.

Reinforcement learning has also been applied to public-service dispatch. Hua and Zaman [6] formulated ambulance

dispatch as an average-cost Markov decision process and developed a temporal-difference policy that outperformed a myopic baseline. Repasky et al. [7] proposed a multi-agent RL framework for joint police patrol and dispatch decisions. However, these studies primarily target emergency response times rather than persistent municipal service backlogs.

Prediction has also been combined with dynamic resource scheduling. Ruan et al. [8] integrated multi-step crowd-flow prediction with an energy-adaptive scheduling heuristic for mobile public resources. While demonstrating the value of anticipated demand, their framework addresses crowd coverage rather than unresolved service requests.

The proposed framework extends this literature by incorporating multi-day, area-level forecasts directly into the PPO observation space and modeling repeated daily allocation under sustained capacity constraints. Because unresolved requests carry forward and age, the policy must balance request resolution, backlog magnitude and age, unused capacity, and crew travel costs.

III. PROPOSED APPROACH

A. Framework Overview

The proposed framework integrates demand forecasting and reinforcement learning in two phases. First, historical 311 requests are aggregated into area-level daily demand and used to train an XGBoost model. The model generates a five-day forecast for each of Vancouver’s 22 local areas, covering the current day and the following four days.

Second, the forecast window is combined with the current backlog, backlog age, and crew locations to form the PPO observation. The agent produces a continuous priority score for each zone, and the highest-ranked zones are selected, with one crew assigned to each.

After each allocation, the environment generates new requests, applies service capacity, and updates the backlog, backlog age, and crew locations. The reward reflects resolved requests, unused capacity, remaining backlog, backlog age, and travel cost. Repeated interaction with the environment enables PPO to learn a forecast-aware allocation policy that balances current conditions with anticipated demand.

B. Data Preparation

The study uses the City of Vancouver 311 service-request dataset [3]. Request creation date and local area were retained because the framework operates at the area level. Address and coordinate attributes were excluded because of substantial missing values, and records without a valid creation date or local-area label were removed.

Creation dates and area labels were standardized, and requests were aggregated by date and local area to obtain daily counts for Vancouver’s 22 service zones. A complete date–area panel was then constructed, with missing combinations assigned a count of zero. The panel was divided chronologically: observations before January 1, 2026 were used to train the forecasting models, while those from January 1 to May 15, 2026 were reserved for out-of-sample forecasting and the RL

environment. All lagged and rolling features were computed using only information available before each prediction date to prevent temporal leakage.

C. Phase I: Demand Forecasting

The first phase estimates the expected daily number of service requests for each local area. Let $y_{z,t}$ denote the number of requests recorded in service zone z on day t . A single forecasting model was trained across all 22 zones to learn shared temporal patterns while retaining area-level differences.

1) *Feature Engineering*: Four categories of predictors were generated from the daily date–area panel: lagged demand, rolling statistics, calendar variables, and spatial identifiers. The lag variables capture short-term and weekly dependencies, while the rolling means represent recent demand trends. Calendar variables account for recurring weekly and seasonal patterns, and the encoded local-area identifier enables the model to distinguish among service zones.

Table I summarizes the predictors used by the forecasting models.

TABLE I
ENGINEERED FEATURES USED FOR DEMAND FORECASTING

Feature	Description
lag_1	Request count from the preceding day.
lag_7	Request count from seven days earlier.
lag_14	Request count from fourteen days earlier.
rolling_mean_7	Mean request count over the preceding seven days.
rolling_mean_30	Mean request count over the preceding thirty days.
day_of_week	Day of the week associated with the prediction date.
month	Month associated with the prediction date.
weekend	Binary indicator identifying weekend dates.
area_encoded	Encoded identifier for the local service area.

For a rolling window of length w , the rolling-mean feature for zone z on day t was calculated as

$$\bar{y}_z, t^{(w)} = \frac{1}{w} \sum_{j=1}^w y_{z, t-j}, \quad w \in 7, 30. \quad (1)$$

The use of observations from $t - 1$ and earlier ensures that the current prediction target is not included in its own predictors.

2) *Forecasting Models*: A Random Forest regression model [9] was implemented as the forecasting baseline. Random Forest combines predictions from multiple decision trees trained on bootstrapped samples, reducing the variance associated with an individual tree. The proposed forecasting component uses XGBoost [4], which constructs trees sequentially so that each new tree reduces the residual errors of the existing ensemble.

Both models were trained using the same engineered predictors and chronological data partition. Their out-of-sample performance was evaluated using mean absolute error (MAE), root mean squared error (RMSE), and the coefficient of determination (R^2). The model with the stronger forecasting performance was selected to generate the multi-day demand

forecasts supplied to the reinforcement learning environment. For each allocation date, the forecast window was generated using only information available at the corresponding forecast origin. ““latex

D. Phase II: Proactive Crew Allocation

1) *Reinforcement Learning Formulation*: Crew allocation is formulated as a Markov decision process with one decision per simulated day. The agent observes predicted demand and the current operating state, assigns crews to service zones, and receives a reward based on request resolution, backlog magnitude and age, unused capacity, and crew travel.

The policy is trained using Proximal Policy Optimization (PPO) [5], an on-policy actor–critic algorithm that uses a clipped surrogate objective to limit destabilizing policy updates. PPO is applied to the continuous priority-vector action space defined below.

2) *Simulation Environment and Observation Space*: The Gymnasium environment [10] models 11 homogeneous crews operating across Vancouver’s 22 local areas. Each crew can resolve up to 30 requests per day, giving a maximum nominal capacity of 330 requests. Each episode spans 30 consecutive days.

At day t , the agent receives a 165-dimensional observation comprising (i) a 5×22 XGBoost forecast matrix for the current and following four days, (ii) 22 log-transformed backlog values $\log(1 + b_{z,t})$, (iii) 22 backlog-age values $a_{z,t}$, and (iv) the locations of the 11 crews. Here, $a_{z,t}$ is the number of consecutive days for which zone z has retained unresolved requests. The logarithmic transformation limits the influence of large backlogs, while the forecast window provides the proactive signal.

At each training reset, a starting date is sampled from the chronological RL-training partition. Crew locations are randomized, and each zone receives an initial backlog sampled uniformly from 0 to 10 requests.

3) *Priority-Vector Action Space*: Allowing each crew to select one of 22 zones independently produces $22^{11} \approx 5.84 \times 10^{14}$ possible labeled assignments. To reduce this complexity, PPO outputs a continuous priority vector $\mathbf{p}_t \in \mathbb{R}^{22}$. The 11 highest-ranked zones are selected as

$$\mathcal{Z}_t^* = \arg \max_{S \subseteq \mathcal{Z}, |S|=11} \sum_{z \in S} p_{z,t}. \quad (2)$$

One crew is assigned to each selected zone. Thus, this formulation imposes distinct-zone coverage and represents a structured version of the unrestricted multi-crew assignment problem.

The selected zones are matched to crews by minimizing total travel cost:

$$\pi_t^* = \arg \min_{\pi \in \Pi_t} \sum_{i=1}^{11} \mathbf{T}[\ell_{i,t-1}, \mathcal{Z}_{\pi(i),t}], \quad (3)$$

where Π_t is the set of one-to-one crew–zone matchings, $\ell_{i,t-1}$ is crew i ’s previous location, and $\mathbf{T}$ is the travel-cost matrix.

4) *Environment Transitions*: For zone z on day t , simulated demand is generated from the XGBoost forecast $\hat{d}_{z,t}$ as

$$d_{z,t} = \max\left(0, \text{round}(\hat{d}_{z,t} + \epsilon_{z,t})\right), \quad \epsilon_{z,t} \sim \mathcal{N}(0, 5^2). \quad (4)$$

Let $q_{z,t}$ denote the pre-service workload, $n_{z,t}$ the number of assigned crews, $s_{z,t}$ the service capacity, and $r_{z,t}^{\text{res}}$ the resolved requests. The daily transition is

$$q_{z,t} = b_{z,t} + d_{z,t}, \quad s_{z,t} = 30n_{z,t}, \quad (5)$$

$$r_{z,t}^{\text{res}} = \min(q_{z,t}, s_{z,t}), \quad b_{z,t+1} = \max(q_{z,t} - s_{z,t}, 0). \quad (6)$$

Because one crew is assigned to each selected zone, $n_{z,t} \in \{0, 1\}$. Backlog age is updated as

$$a_{z,t+1} = \begin{cases} a_{z,t} + 1, & b_{z,t+1} > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

This variable measures zone-level backlog persistence, not the waiting time of individual requests.

5) *Reward Function*: The daily reward is

$$r_t = 0.1R_t - 0.1W_t - 0.05\bar{B}_t - 0.4A_t - 0.2C_t, \quad (8)$$

with components defined as

$$R_t = \sum_{z=1}^{22} r_{z,t}^{\text{res}}, \quad (9)$$

$$W_t = \sum_{z=1}^{22} \max(s_{z,t} - q_{z,t}, 0), \quad (10)$$

$$\bar{B}_t = \frac{1}{22} \sum_{z=1}^{22} b_{z,t+1}, \quad (11)$$

$$A_t = \frac{1}{22} \sum_{z=1}^{22} \log(1 + b_{z,t+1})a_{z,t+1}, \quad (12)$$

$$C_t = \sum_{i=1}^{11} \mathbf{T}[\ell_{i,t-1}, \ell_{i,t}]. \quad (13)$$

The travel matrix groups the 22 areas into Downtown Core, East Vancouver, Central Vancouver, and West Side, with lower costs for within-region movement than for movement between regions. Daily crew utilization is calculated as

$$U_t = \frac{R_t}{\sum_{z=1}^{22} s_{z,t}}. \quad (14)$$

6) *PPO Training Configuration*: PPO was implemented using Stable-Baselines3 [11] and trained for 1,000,000 timesteps. The initial 600,000-step phase used the age penalty $b_{z,t+1}a_{z,t+1}$. Training then continued for 400,000 steps using the final log-compressed term $\log(1 + b_{z,t+1})a_{z,t+1}$, which reduced the influence of large backlogs on the reward.

Checkpoints were saved every 50,000 timesteps and evaluated every 25,000 timesteps using deterministic actions in

a separate validation environment. The checkpoint with the highest mean validation reward was selected for final testing, which used a separate chronological test partition.

TABLE II
PPO HYPERPARAMETERS

Hyperparameter	Value
Total timesteps	1,000,000
n_steps	512
Batch size	128
Learning rate	3×10^{-4}
Discount factor (γ)	0.99
Entropy coefficient	0.01
Reward normalization	VecNormalize
Reward clipping	10

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Dataset Characteristics and Exploratory Analysis

The final dataset contained approximately 1.18 million municipal service-request records distributed across Vancouver’s 22 local areas. Request volumes were not uniformly distributed across the city. As shown in Figure 1, Downtown recorded the highest number of requests, followed by Kensington-Cedar Cottage, Kitsilano, Renfrew-Collingwood, and Grandview-Woodland. In contrast, South Cambie, Shaughnessy, and Arbutus Ridge recorded substantially lower request volumes.

This spatial variation supports the use of area-level demand forecasting because a single city-wide estimate would not represent the distinct demand patterns observed across service zones. The values in Figure 1 represent total request counts and are not normalized by population or geographic area.

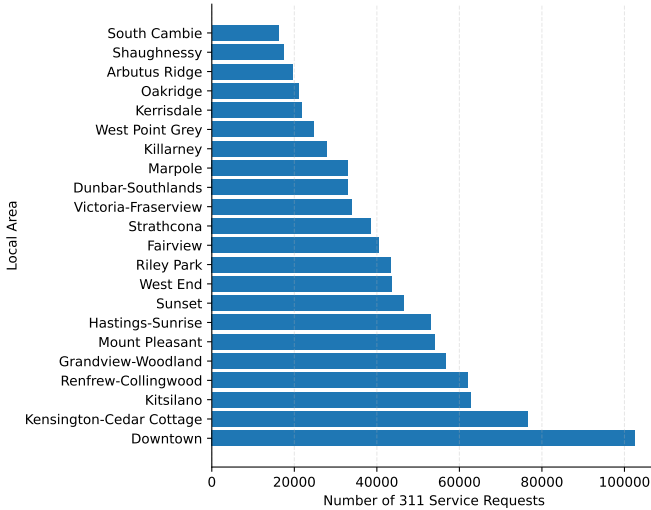


Fig. 1. Total number of 311 service requests by Vancouver local area during the study period.

Temporal analysis also revealed systematic variation in demand across the days of the week. Figure 2 presents the average daily request volume for each weekday. The observed weekly variation supports the inclusion of calendar variables,

including day of the week and the weekend indicator, in the demand-forecasting models.

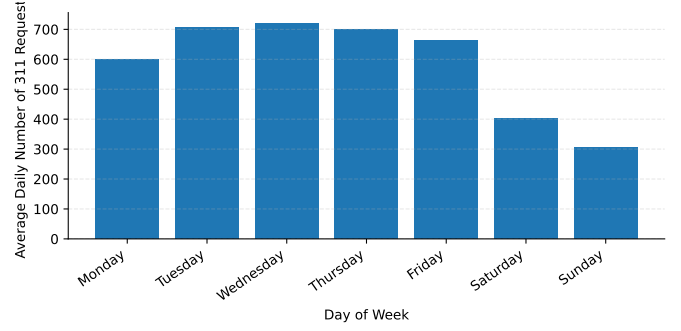


Fig. 2. Average daily 311 service-request volume by day of the week.

B. Demand-Forecasting Results

Random Forest and XGBoost were evaluated on the chronological forecasting partition using mean absolute error (MAE), root mean squared error (RMSE), and R^2 . The results are reported in Table III.

TABLE III
DEMAND-FORECASTING PERFORMANCE

Model	MAE	RMSE	R^2
Random Forest	7.90	62.88	0.760
XGBoost	7.89	62.30	0.763

XGBoost provided a modest improvement across all three metrics and was therefore selected to generate the forecasts used by the RL environment. The substantially larger RMSE than MAE indicates that a limited number of observations produced comparatively large errors. As shown in Figure 3, the model nevertheless captured the general variation in daily demand. Feature-importance scores further indicated that lagged demand and rolling averages were the dominant predictors, with calendar and area variables providing additional differentiation.

C. Crew-Allocation Evaluation

1) *Comparison Policies*: The proposed PPO policy was evaluated against three comparison policies. All policies operated under the same simulated demand, initial backlog, initial crew locations, and service-capacity conditions.

Magnitude-Based Greedy: The policy selected the 11 zones with the highest combined current backlog and predicted current-day demand. It did not consider backlog age, crew travel cost, or demand beyond the current day.

Reactive FCFS: Crews remained in their current zones while unresolved requests were present. When a crew became idle, it was reassigned to a zone containing the longest-waiting unresolved requests.

Oldest-or-Closest: Zones were prioritized according to the age of their unresolved backlogs. Travel distance from the current crew location was used to break ties, and duplicate zone assignments were not permitted.

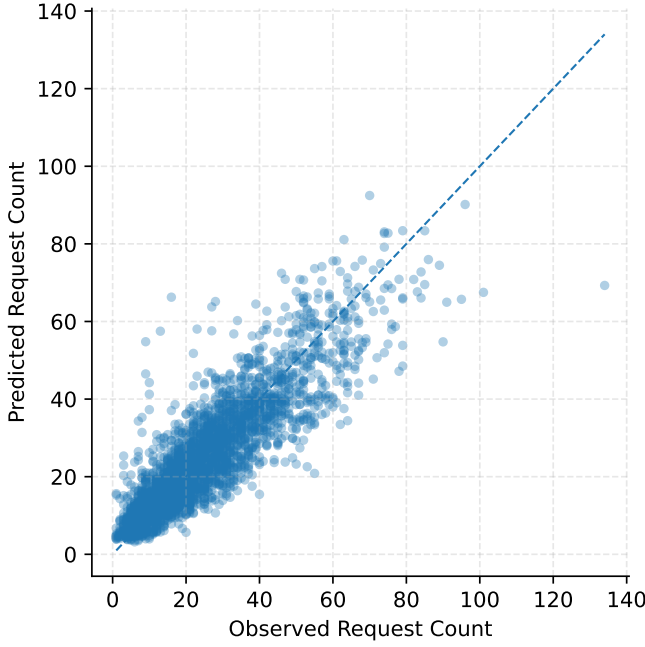


Fig. 3. Observed and XGBoost-predicted daily request volumes during the forecasting evaluation period.

2) *Evaluation Protocol*: Following model training and checkpoint selection on a validation partition, all policies were evaluated on 100 paired simulation episodes drawn from a separate chronological test partition. Each episode represented 30 consecutive days. Identical random seeds were used for all policies so that they encountered the same starting conditions and stochastic demand realizations.

No test episode was used during PPO training or checkpoint selection. The rewards reported in this section are raw, unnormalized episode rewards, allowing direct comparison across the four policies.

3) *Training Behavior*: Figure 4 shows the mean episode reward observed during PPO training. The policy demonstrated an improving reward trend as training progressed, indicating that it learned to make more effective allocation decisions within the simulated environment.

Because the age-pressure component of the reward was modified between the two training phases, the absolute reward values before and after the phase boundary are not strictly comparable. The figure should therefore be interpreted as an illustration of optimization progress within each phase rather than as a single-objective convergence curve.

4) *Policy Comparison*: Table IV reports the mean and standard deviation of episode rewards. PPO achieved the highest mean reward and lowest variability, exceeding Oldest-or-Closest, the strongest baseline, by 33.7 points and Magnitude-Greedy by 65.5 points. Reactive FCFS produced the lowest mean and greatest variability, indicating that its limited repositioning behavior was poorly suited to sustained demand across multiple zones. The distributions in Figure 5 similarly

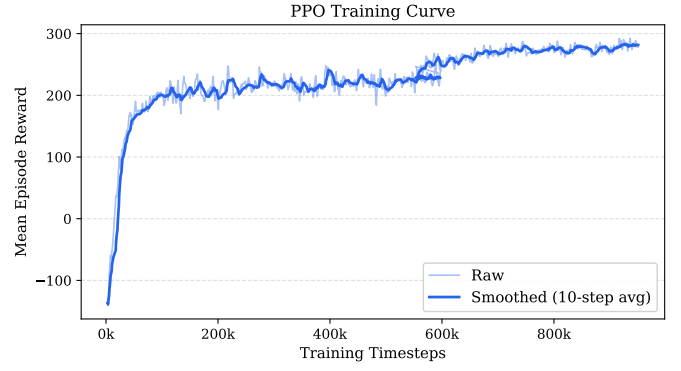


Fig. 4. Mean episode reward during PPO training across the two training phases.

TABLE IV
POLICY COMPARISON ON THE TEST PARTITION

Policy	Mean	Std.	Gap
PPO (Proposed)	262.7	50.5	–
Oldest-or-Closest	229.0	67.0	–33.7
Magnitude-Greedy	197.2	70.2	–65.5
Reactive FCFS	–700.8	167.9	–963.5

show PPO with the highest median and narrowest interquartile range.

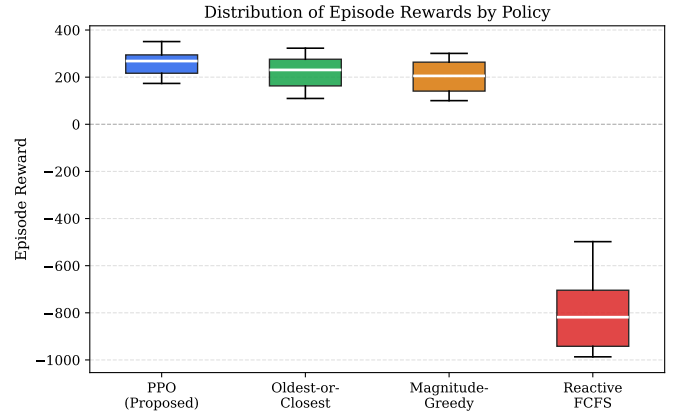


Fig. 5. Raw episode-reward distributions across paired test episodes.

D. Backlog and Capacity Analysis

Predicted demand typically ranged from 500 to 700 requests per day, while the 11 crews provided a maximum nominal capacity of 330. Accordingly, backlog increased under every policy. Figure 6 shows that PPO maintained the lowest trajectory, particularly later in the episodes as the effects of allocation decisions accumulated. Because demand and nominal capacity were fixed across policies, the lower backlog suggests that PPO directed crews toward zones with sufficient workload and avoided more unused capacity.

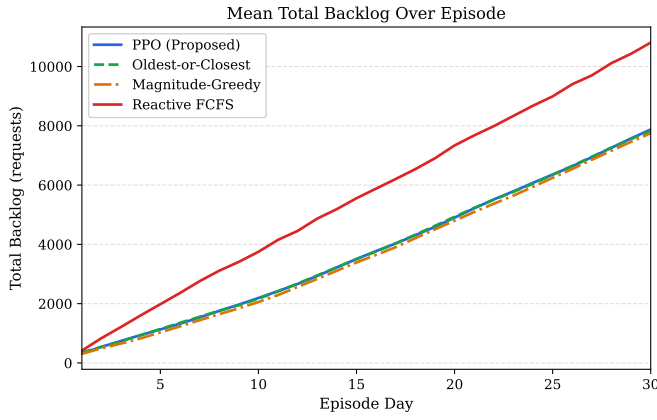


Fig. 6. Mean total backlog by episode day; shaded regions show one standard deviation.

E. Discussion

The evaluation results suggest that combining demand forecasts with current operational information can improve daily crew allocation under persistent capacity constraints. The PPO policy used anticipated demand, current backlog, backlog age, and crew locations to produce a dynamic priority score for each zone. Its performance relative to Oldest-or-Closest suggests that a learned policy can capture interactions among these variables that are difficult to represent using a fixed priority rule.

The continuous priority-vector formulation also reduced the complexity of the policy output. Rather than independently selecting a destination for every crew, the policy ranked the 22 service zones and selected 11 distinct destinations. This formulation made PPO training more tractable but also imposed a maximum of one crew per selected zone. It should therefore be interpreted as a structured allocation model rather than an equivalent reparameterization of unrestricted multi-crew assignment.

The policy comparison does not, by itself, completely separate the benefit of reinforcement learning from the benefit of receiving a multi-day forecast. PPO had access to the complete forecast window, whereas the comparison policies used either current conditions or current-day predicted demand. A forecast-aware heuristic using the same multi-day information would provide a stronger comparison and help quantify the contribution of the learned policy.

V. CONCLUSION

This paper presented a simulation-based, two-stage framework for proactive municipal service allocation. The first stage uses XGBoost to forecast area-level 311 service demand, while the second stage uses a PPO agent to combine multi-day forecasts with current backlog, backlog age, and crew-location information. A continuous priority-vector formulation was introduced to make the allocation problem computationally tractable by ranking service zones and selecting distinct destinations for the available crews.

Across the paired simulation episodes, PPO achieved the highest mean reward and the lowest reward variability among the evaluated policies. It also maintained a lower backlog trajectory under conditions in which daily demand exceeded available service capacity. Within the assumptions of the simulation, these findings suggest that combining demand forecasting with reinforcement learning can support more proactive and consistent municipal resource-allocation decisions.

Future work will incorporate actual staffing, service-time, and road-network travel data; heterogeneous request types and crew capabilities; stochastic crew availability; and more realistic models of forecast uncertainty. Additional evaluation should include forecast-aware heuristic baselines, multiple independent PPO training runs, reward-weight sensitivity analysis, and validation across other cities and municipal service domains.

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